

GME Stock Price Prediction & Sentiment Analysis

Executive Evaluation Report — Machine Learning Phase 8

Project Executive Summary

This predictive modeling framework combines historical financial market data for GameStop (GME) with alternative data scraped from social media (Reddit). By lagging social indicators strictly by 24 hours, the model simulates a production environment designed to forecast daily price direction before the market opening bell.

Model Performance Metrics

The constrained Random Forest Classifier successfully broke past the standard 50.00% random walk baseline, achieving a final historical testing directional **accuracy score of 66.67%** on unseen data. This performance tier strongly indicates a statistically significant predictive alpha signal.

Metric Classification	Baseline Score	Optimized Model Score
Directional Accuracy	50.00%	66.67%
Data Timeline Focus	29 Trading Days	29 Trading Days
Top Predictive Driver	N/A	Yesterday_Posts (Hype Volume)

Core Analytical Insight

Feature evaluation indicates that **Yesterday_Posts** dominated model importance constraints at over 25%. This confirms that retail attention metrics act as an explicit leading market momentum indicator for highly volatile equity structures, proving heavily decoupled from underlying fundamental values.